Rio Ben-Yehoshua

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EDUCATION

The Cooper Union for the Advancement of Science and Art, New York, NY *Expected May 2029*
Bachelor of Engineering in Mechanical Engineering | **GPA: 3.77**
Half-Tuition Scholarship 2025-2029 | Dean's List
Relevant Coursework: Engineering Design, CAD, Statics, Dynamics, Physics I & II, Multivariable Calculus, Differential Equations, Materials Science, Physics & Chemistry Lab

PROJECTS

Ping Pong Smart Hoop – Velocity tracking hoop with launcher and return system *March - May 2026*
- Designed and laser-cut the hoop in Onshape CAD, evolving it from an IR sensor enclosure to final Angry Birds structure
- Designed and 3D-printed a two-wheel flywheel launcher, iterating mounts and wheel grip to survive vibration
- Built a bounce-back return with a slack catch net and reinforcing panels, recirculating 6 balls for 60-second runs
- Wrote Arduino control code for L298N flywheel and MOSFET funnel motors, later extended by a teammate on ESP32
- Prototyped an 8-sensor IR hoop (**±5% velocity accuracy**) before the team switched to camera-based counting
- Collaborated in a 4-member team to deliver an autonomous system that placed **1st in class with 110 passes** in 60 sec

Digital Load Cell Scale with LCD Interface *April - May 2026*
- Designed and CAD-modeled a compact enclosure integrating load cell, LCD, controls, wiring, and electronics
- Programmed an Arduino-based weighing system with HX711 sensing, tare/reset, unit switching, and I2C LCD output
- Achieved **0.84% full-scale error** using 25-sample averaging, 1-second LCD updates, and a 0.05 g zero threshold

PlantPlex – Modular Indoor Urban Farming System (without pumps) *September - December 2025*
- Designed and prototyped a modular vertical farming system sized to fit standard apartment dimensions
- Built a passive capillary-action irrigation system sustaining consistent soil moisture across all pods without pumps
- Selected and validated a four-crop system meeting >50% daily value for priority micronutrients per adult
- Contributed to introduction, problem scoping, and iterative prototype development across a 7-member team

LEADERSHIP & ACTIVITIES

Motorsports Club - The Cooper Union *Spring 2026 - Current*
Body Cockpit Safety Team Member
- Evaluated harness ergonomics and mounting methods; designed a pedal box chassis mount in SolidWorks
- Researched suspension architectures and spring-damper specifications to support front and rear suspension maintenance
- Reviewed FSAE requirements and evaluated materials for floor closeout, firewall, and thermal insulation systems

Krav Maga - Steve Sohn's Krav Maga Training Center *September 2016 - September 2025*
Student and Volunteer
- Achieved advanced rank through 9+ years of disciplined weekday practice, developing situational awareness, resilience, and tactical problem-solving
- Volunteer Assistant Coach: Mentored students ages 11–70, adapting drills and providing technical corrections to help beginners, advanced students, and autistic students build confidence and skill

Organizations: Sophomore Representative in ASME, Society of Women Engineers Member, Hillel Member

PROFESSIONAL EXPERIENCE

Playa Bowls, Rye, NY *Fall 2023 - Summer 2026*
Shift Leader
- Led a team of 3–6, managing task delegation and onboarding 5+ new employees
- Managed daily financial reconciliation including register counts, cash deposits, and safe audits with full accuracy

SKILLS

Programming: C, Python, Arduino,
CAD & Technical: Solidworks, Autodesk Fusion, Onshape, LaTeX, Microsoft Excel
Fabrication & Prototyping: 3D printing, laser cutting, soldering, band saw, table saw, machine shop certified
Languages: English (Fluent), Hebrew (Fluent), Chinese (Conversational)